COMP1589 - Week 2 labs

This work is part of your assessment and when finished should be saved in a folder called “week2_lab”, this will then be zipped and uploaded to the coursework upload slot.

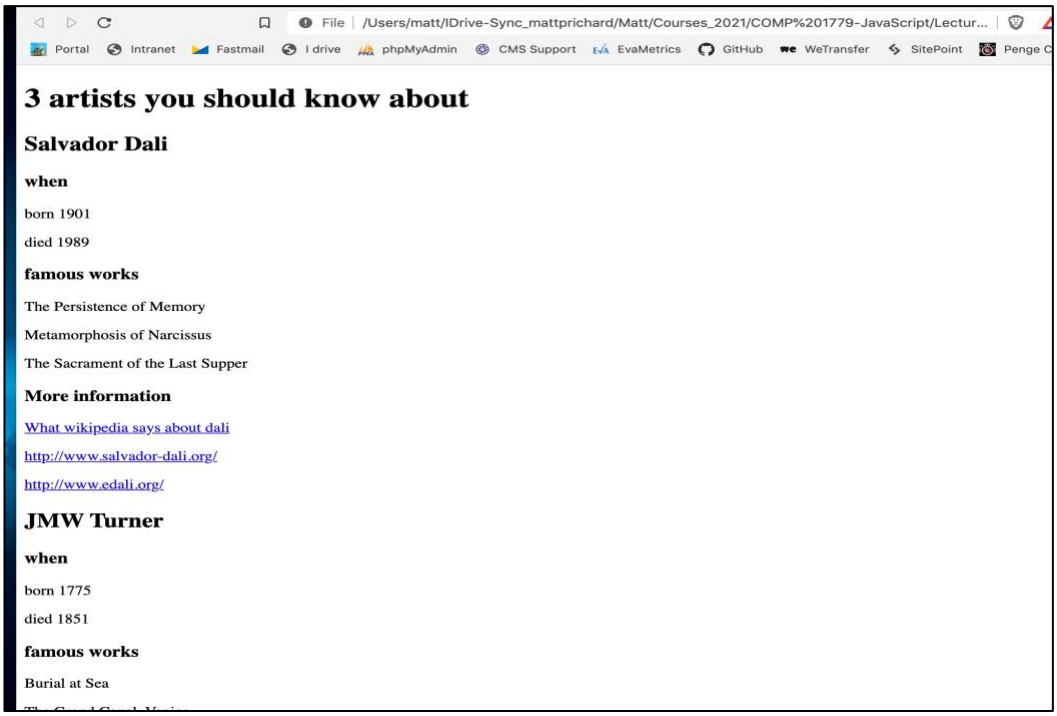
So far we have concentrated on marking up content to the HTML5 standard. The display relies on the default formatting of the browser. You can override those defaults by using style sheets.

There are three ways, which can be combined, to create styles:

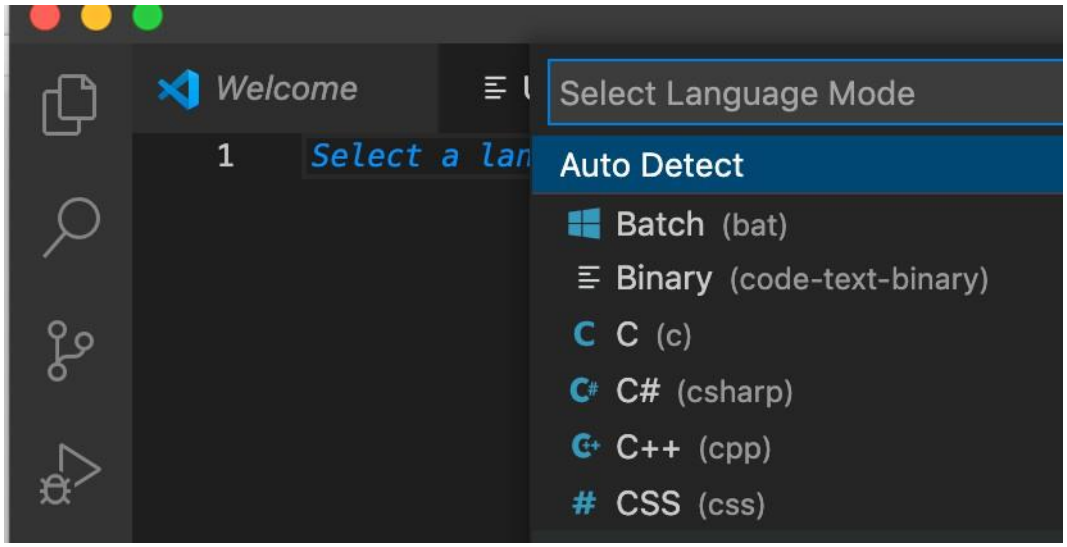
- A style attribute in a tag (not recommended)
- A style element in the head element of a web page.
- A style sheet file that is linked to a web page (recommended)

Preparatory work

Download the “week2_lab” zip file from today’s Moodle page, unzip it to your COMP1589/week2 folder on your G Drive or One Drive
If you view the week2_task1.html file in your browser, you should see this.

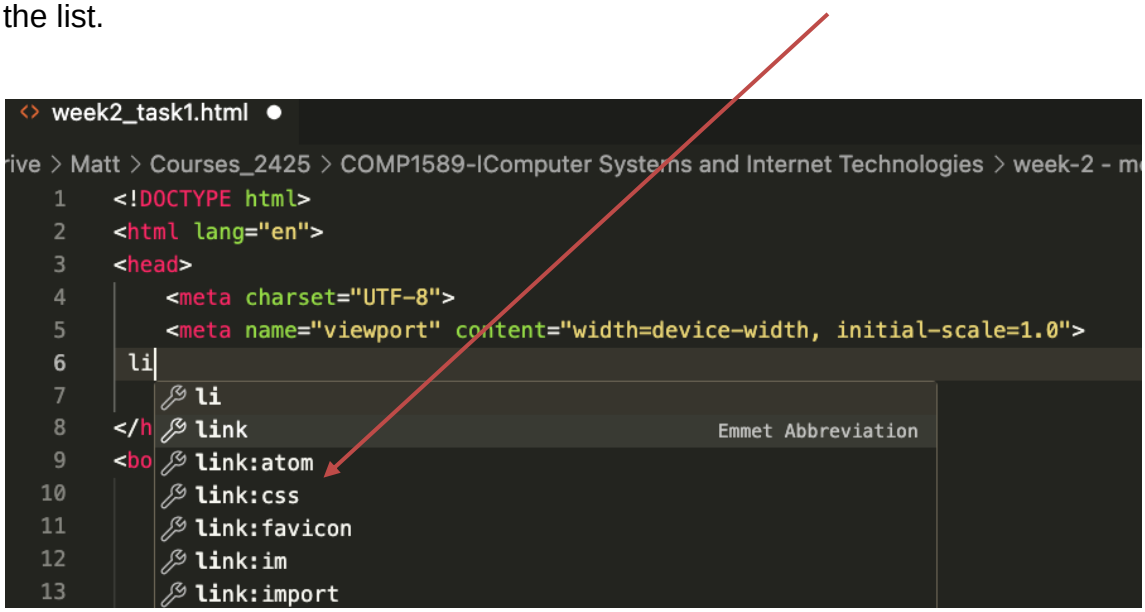


In VS Code create a new file and set the language to CSS like we did for HTML last week

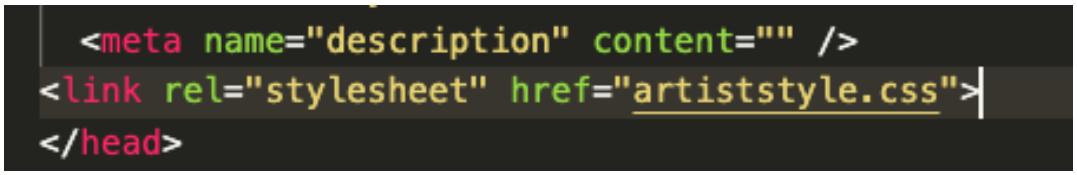


Save this file as *artiststyle.css* in the same folder as your *week2_task1.html* web page.

Open your *week2_task1.html* web page in VS code, make a new line at line 6 and start to type "link" and the auto complete should offer you "link:css" – select this from the list.



Next update the 'href' to match the name of your css page – 'artiststyle.css'



Save your *week2_task1.html* page

Styling elements

Body element

Back in your style sheet, type in this code: (exactly, look at the punctuation very carefully and get it right)

```
body {
background-color: #99bbff;
color: #000000;
font-family: Verdana, Geneva, Tahoma, sans-serif;
}
```

Save this file.
Go to your web page and view it in a browser.

Go back to the style sheet, what do you think the *color* (Notice American spelling) controls. Change its value to *#FF0000* . Save and look at web pages. You can change any of the six digits to any number or any letter A-F. these are hexadecimal codes for colours. Look at https://www.w3schools.com/colors/colors_picker.asp for more information and colour guide.

You have specified the foreground colour (text) and the background colour for all children of the body element that uses this style sheet.

The style specifies the font to use. It is a list because not all viewers may have your first choice. It is pretty standard to have your first choice followed by
`,Arial, Helvetica, sans-serif;`

Or

`,"Times New Roman", Times, serif;`

Additional task in own time
Research how https://fonts.google.com/
or
https://www.w3schools.com/howto/howto_google_fonts.asp
can allow you to get fonts displaying on all devices.

Now suppose you wish your heading 1 element to be in the centre of the window, and larger than the default size.

Heading elements

Add to your style sheet

```
h1 {
    font-size: 3em;
    background-color: #CCCCCC;
    text-align: center;
    }
```

US spelling

Save and preview your web pages.

- `font-size:` means that h1 content is going to be 3 ems in size, we will look at this in more detail later in the course. You can use other measures (e.g 300% or 25pt) but this one is browser friendly.
- `background-color`: you have already met
- `text-align:` can be **left**, **center**, **right**, or **justify**

You can change the text colour for h1, insert on a new line
`color: #550000;`

Change the colour if you don't like it.

Add to your style sheet

```
h2 {
    font-style: italic;
    font-weight: bold;

}
```

you can replace italic or bold by normal, if you want to remove default bold or italic. Change the size and colour of the h2 formatting by using your experience of formatting h1. (copy and paste may help)

Anchor element (styling links)

Now, using the skills you have just learnt, add a style for the anchor element which makes it a colour other than blue.

Save the CSS and refresh and view your page in the browser

Back in the CSS, after code for the anchor element add this bit of code

```
a:hover { background-color: #CCCCCC;}
```

Save the CSS and refresh and view your page in the browser

There's a lot more you can do with the formatting of this element, for the keen look at the link https://www.w3schools.com/css/css_link.asp

Can you work out how to remove the underlining of the link?

Images in html

Suppose you want to include pictures of your artists in the document. Look inside the "images" folder.
Add image tags just under the relevant h2 tag for each artist, referencing the relevant file names. Eg:

```
<h2>Salvador Dali</h2>
<img src="images/dali.jpg" alt="picture of dali as an older man" />
```

NOTE: you **must** have an alt tag that describes the picture briefly. This makes your web page that bit more accessible, helps make your site legal.

NOTE: pictures have been cropped and resized in Photoshop. You should **not** resize in CSS unless know about responsive design.

Save the html and view in browser.

Add images for the other 2 artists

Floating elements (using styles)

Often you want text to appear down the side of a picture.

You can do this by giving a selector the property

```
float: left;
```

Then, when the text has cleared the picture, it will continue to the left of the browser as before.

Add to your style sheet

```
img { float: left;}
```

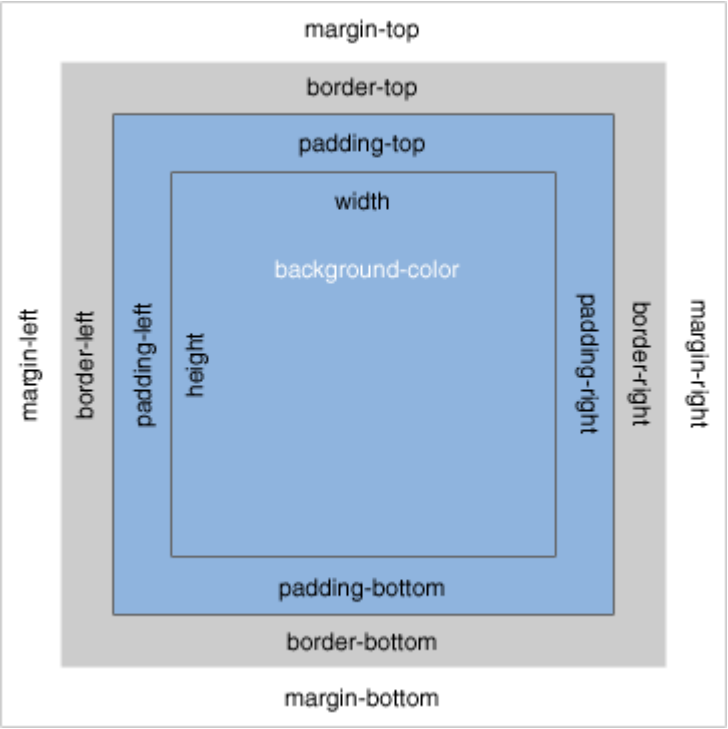
View in your browser, all images will float to the left.

Change left to right and see the effect.
Change back to left

Margins and Padding

Looking at your web page , Have you noticed that there are gaps between the headings and paragraphs. But the text is very close to the picture?

Web pages by default add margins and padding to each element but often in different ways. You have to deal with this!



now add to your img styling

```
img { float: left;
    background-color: #AAFFAA;
    border:solid 5px #005500;
    margin:9px;  padding:9px;
}
```

Now play with the values to see what effect they have on your web page. Decide which is best and choose appropriate values.

Using classes

What if you want images to have different colour borders and backgrounds?

You will need to use what's called a class selector in your style sheet.
e.g. add these two lines after the img styles

```
.redbord { border: solid 5px #550000;}
.blubord { border: solid 5px #000055;}
```

This is called the **selector**; note the dot at the start. This means it is a **class selector**. It can be any name you want but make it sensible

This is called a **property** (of the selector)

This is called a **value** (of the property)

Then back in your HTML file
Change one of the image tags to
`<img class="redbord" src=" ... >`

And another to
or `<img class="blubord" src=" ... >`

Save both files and preview in browser

IMPORTANT: a class in the CSS begins with a dot but NOT in the HTML. The class names must match exactly.

Experiment with other values here:

https://www.w3schools.com/css/css_background.asp

Controlling the width of an object

In your style sheet add the width property to the **body** selector `width: 1000px;`
What effect does this have on your web page? Resize your browser width.

How do you make the page centred in your bowser?

Try this in the body selector (a big mistake but you should try it once never to repeat!)

```
text-align: center;
```

What does this do? Only the text has been centred inside the space available. Look how the wrapped text around pictures has centred.

Recommendations and good practice: only use `text-align: center;` to centre headings (unless you want to look like an amateur).
DO NOT CENTRE BODY TEXT

Delete this code from the body tag

Now add to the body tag `margin-left:auto; margin-right:auto;`
What does this do?

The auto means that the left and right margins (of the body) are automatically adjusted to be the same size. That makes sense.

Summing up the styling of the artist's page

You have started on the road to making things look good using CSS:

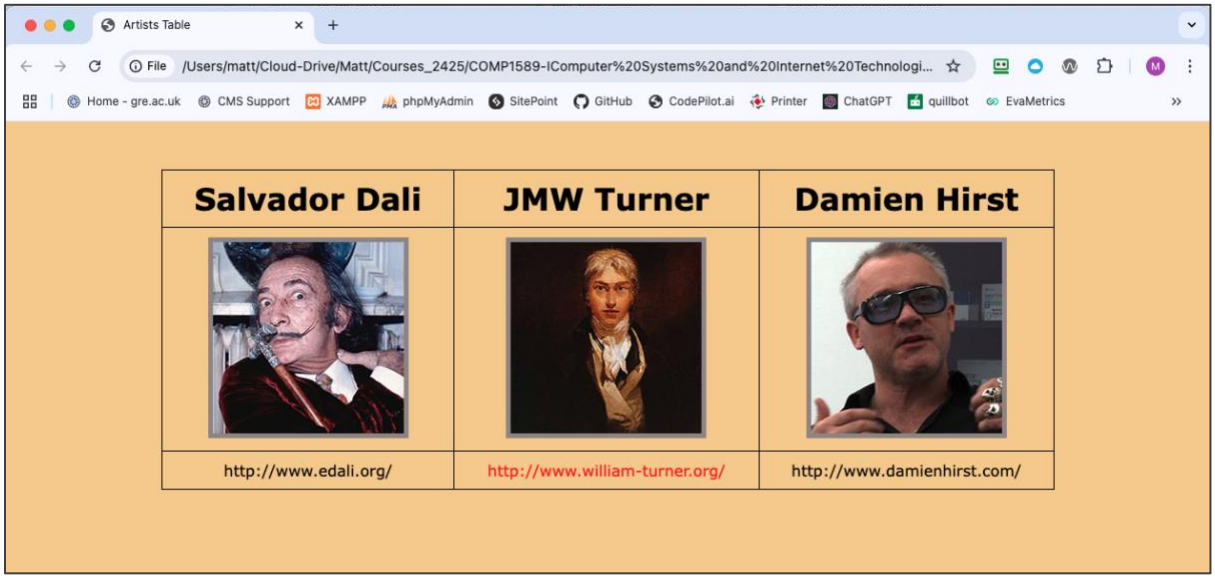
- You used colour
- Centred text (headings only remember)
- Controlled the font appearance
- Floated objects to create word wrap.
- Learnt about margins, padding and borders
- Set the width of objects and learnt how to centre objects

You may end up with something like this Task 1 is now finshed, move on to Task 2.

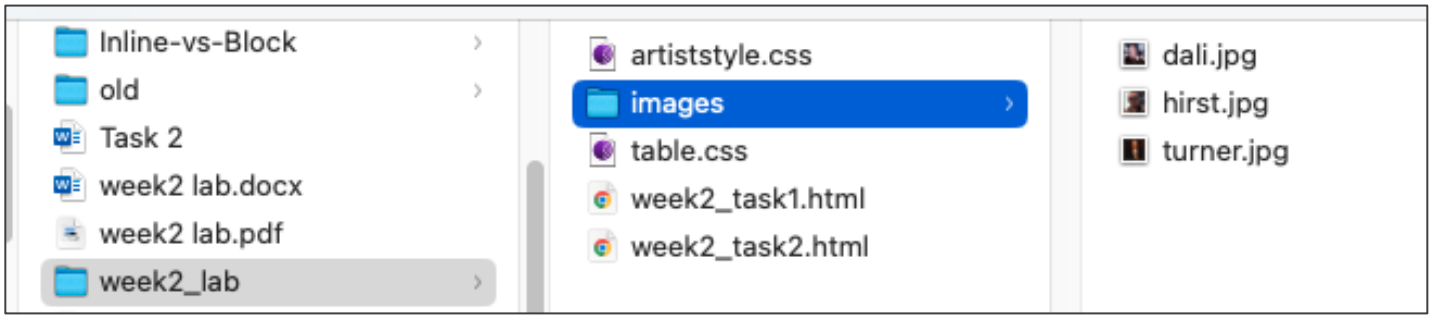
Task 2

In the same folder as the previous task create a new html document in VSCode and save it as week2_task2.html

Below is screen shot of what you need to create. You will be given a list of criteria, and it is up to you to work out how to reproduce them.



You will first need to create and connect a new CSS page called ‘table.css’

1. Basic page set up - background colour as close to the above as you can get, font family Verdana and text colour black.
2. Title the document “Artists Table”.
3. You need a table that is 3 x 3 with the top row being a table header.
4. Table needs to be 75% of the page width.
5. Table need to be 50px from the top of the page and the whole table needs to be centred in the page so that it stays centred if the browser is resized.
6. All table cells need a 1px black border.
7. All cell content needs to be centred, text and images.
8. All content needs 10px padding on all sides
9. Text in the table header needs or be 2ems in size
10. Images need a 5px grey border
11. Links need to be black with no underline and open in a new browser tab.
12. When hovered over links need to turn red as shown above
13. Use the images provided in the previous task.

When complete you will have a folder called week2_lab containing 2 html files and 2 CSS files plus the images folder as shown below. Zip this whole folder up and upload it to the coursework system.